

Data Presentation

Connor J. Cassidy
8/26/2026

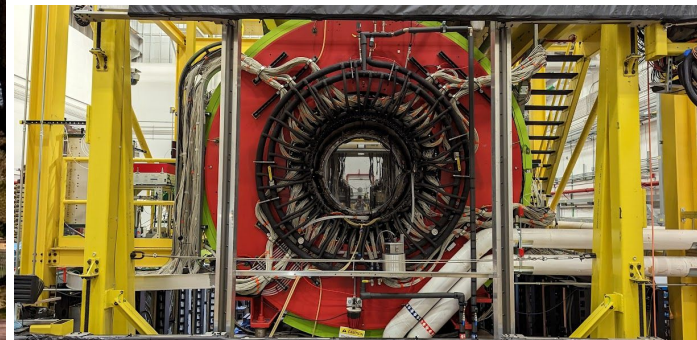
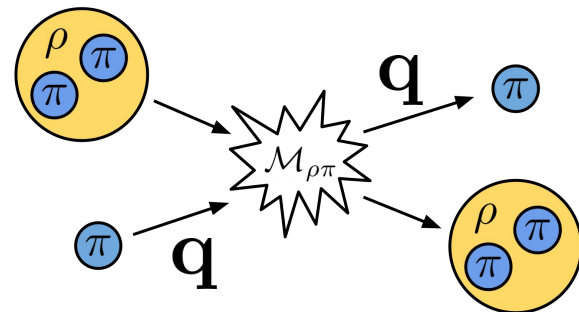
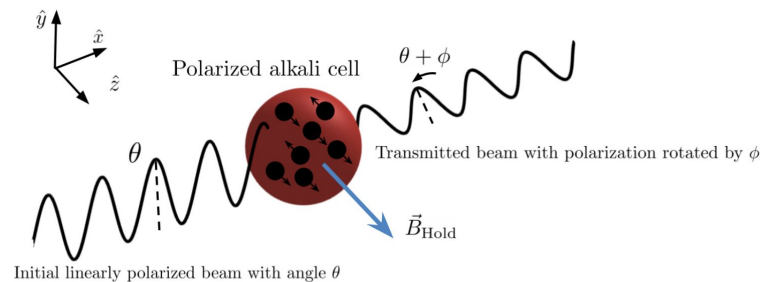
Circularly Polarized
XUV Photoionization
Time Delays

Introduction



WILLIAM
& MARY

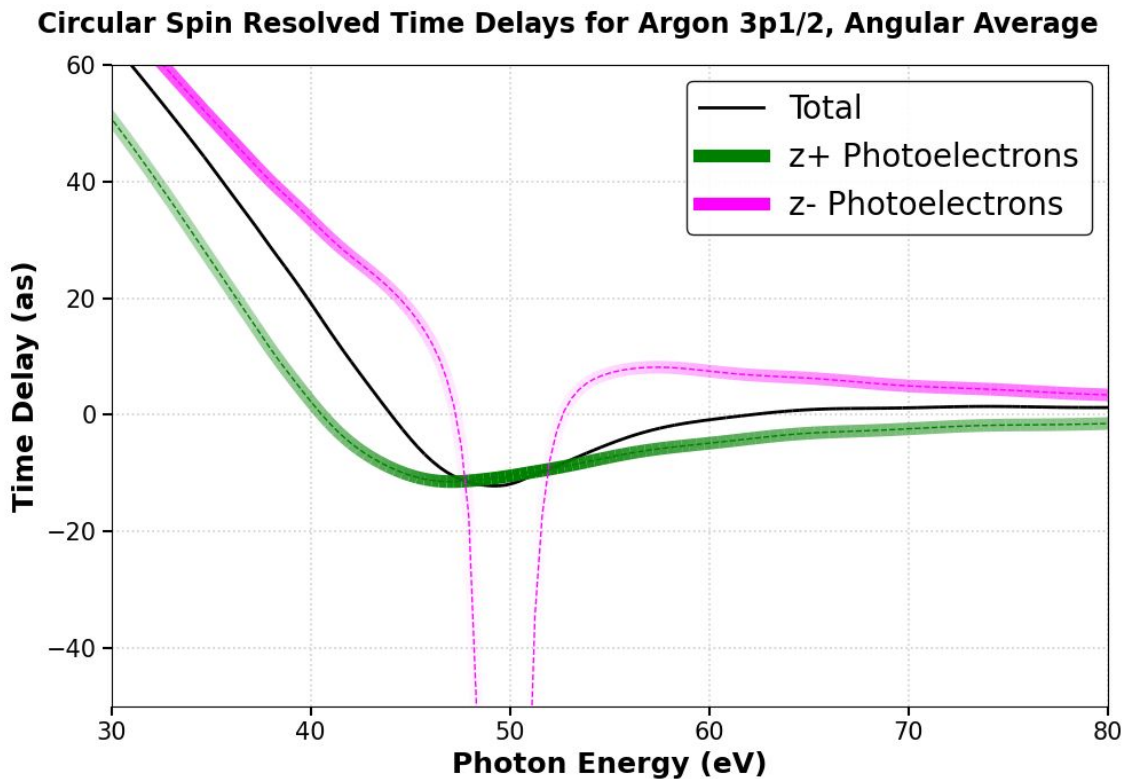
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★ Designed by TownMapsUSA.com

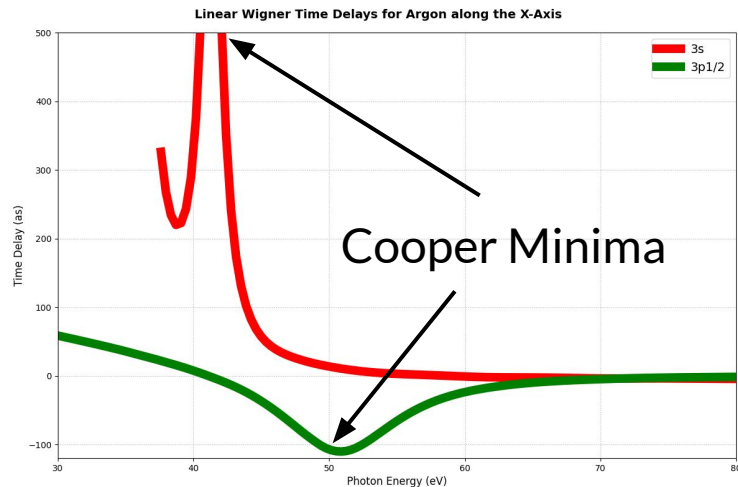
Overview of Research

- Brock had been looking into using RRPA for linear light.
- I started looking into how switching to circular light affects Wigner time delays
- Looked into spin polarization and spin-dependent time delays that are caused by circular light and not present in the linear case



Background

- Wigner Time Delay: Scattering time delay obtained by a particle due to a potential
- Cooper minimum: sign change in dipole matrix elements causes minimum of photoionization cross section and large variation in phase shift
- We will examine time delays around Cooper minima, which is in the XUV range for Ar and Kr.



Without Potential:

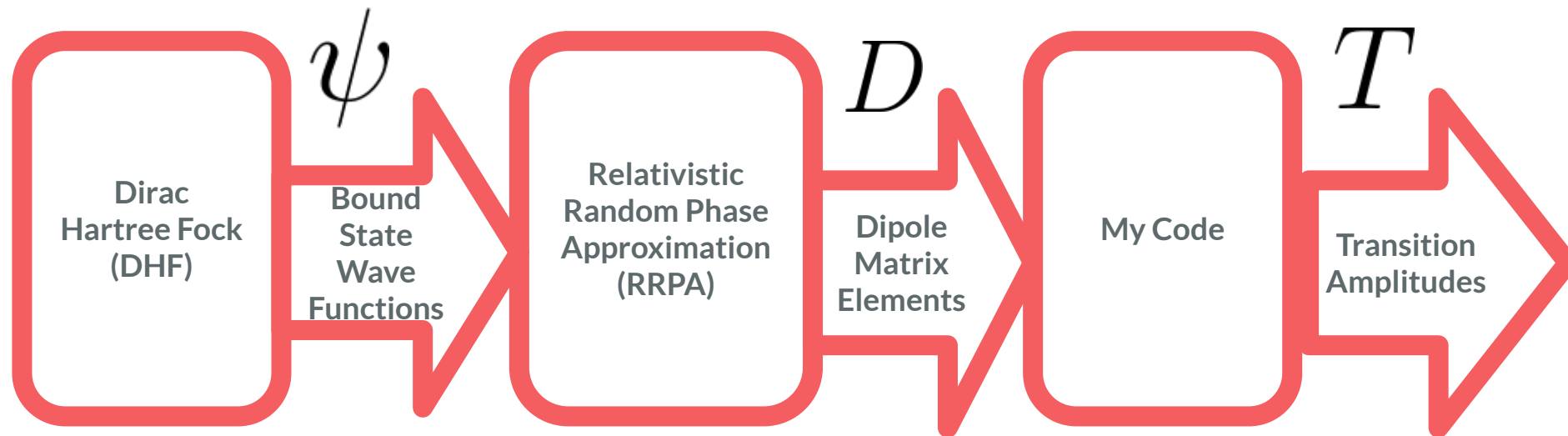


With Potential:



$$\tau_w = t_{\text{potential}} - t_{\text{free}} = \frac{d}{dE} \arg T$$

Methods



W. Johnson, C. Lin, PRA 1979

Transition Amplitudes

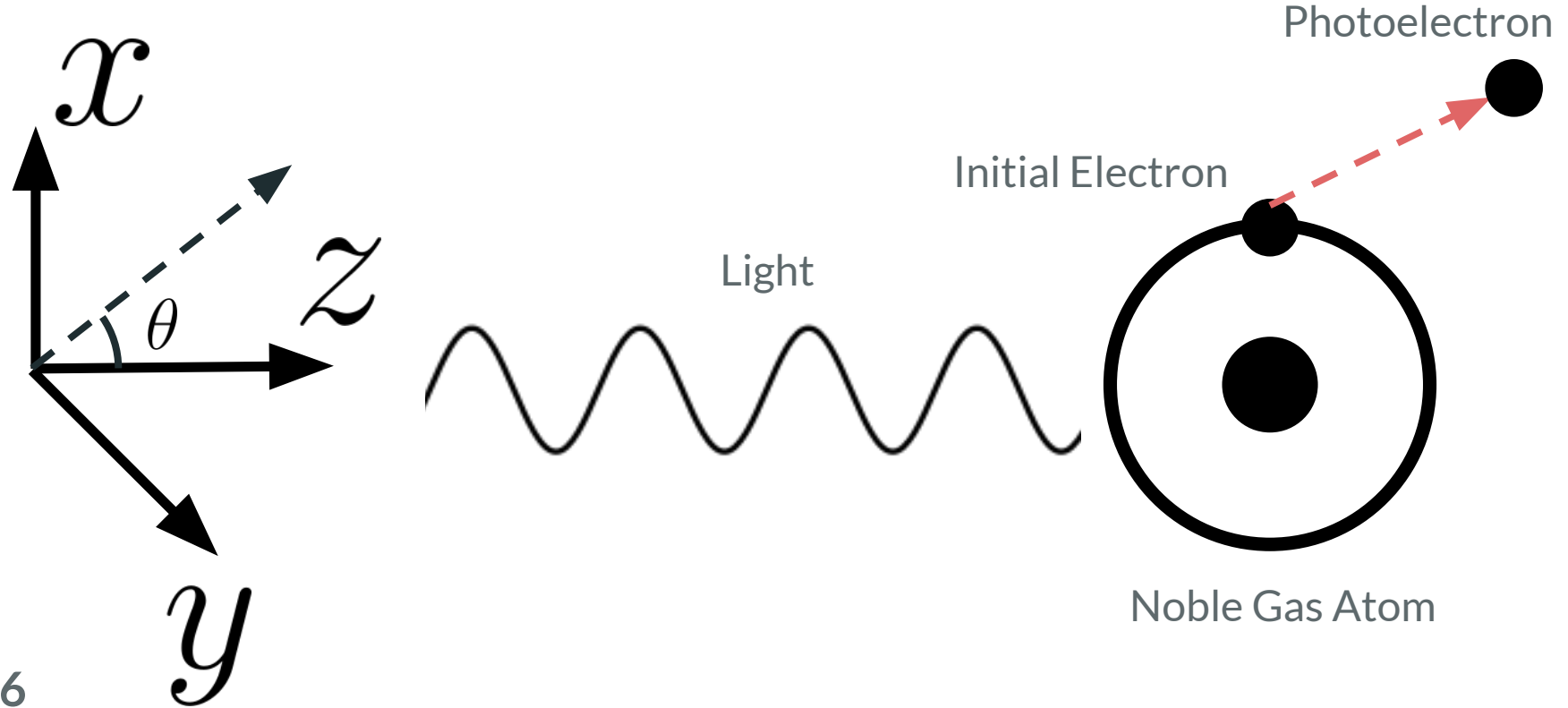
- Amplitudes are linear combinations of dipole matrix elements.
- Fixed initial state, sum over all channels from that initial state
- Cross sections depend on the magnitude of amplitudes, while Wigner time delays depend on their phases.

Amplitudes: $T(E, \theta, \phi) = \sum C_i(\theta, \phi) D_i(E)$

Cross Sections: $\frac{d\sigma}{d\Omega} \propto |T|^2$

Wigner Time Delays: $\tau_W = \frac{d}{dE} \arg T$

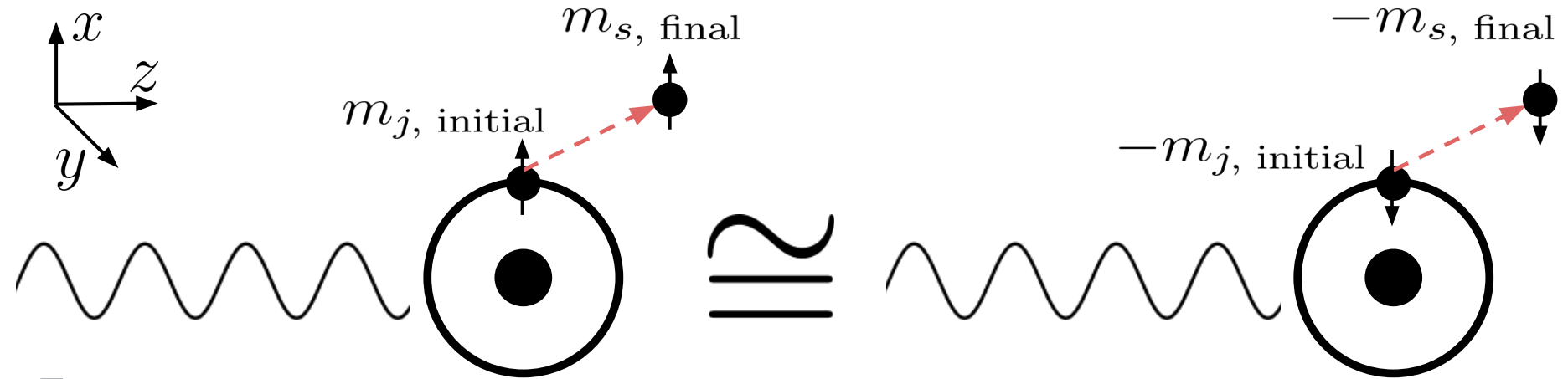
Scattering Coordinate System



Linearly Polarized Light

A. Kheifets et al. PRA 2016

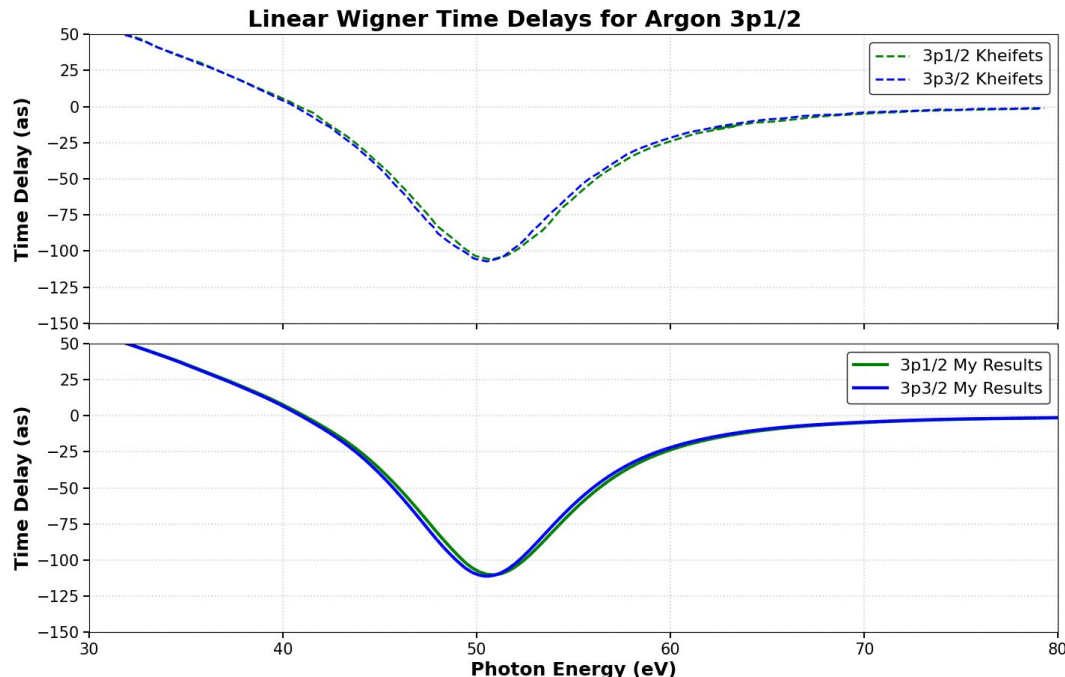
- E-field is symmetric about the x-axis
- Amplitudes have the symmetry shown below, up to a phase.
- The symmetry implies that when spin-resolving along the x-axis, there will be no differences in time delays or spin polarization.



Comparison with Kheifets

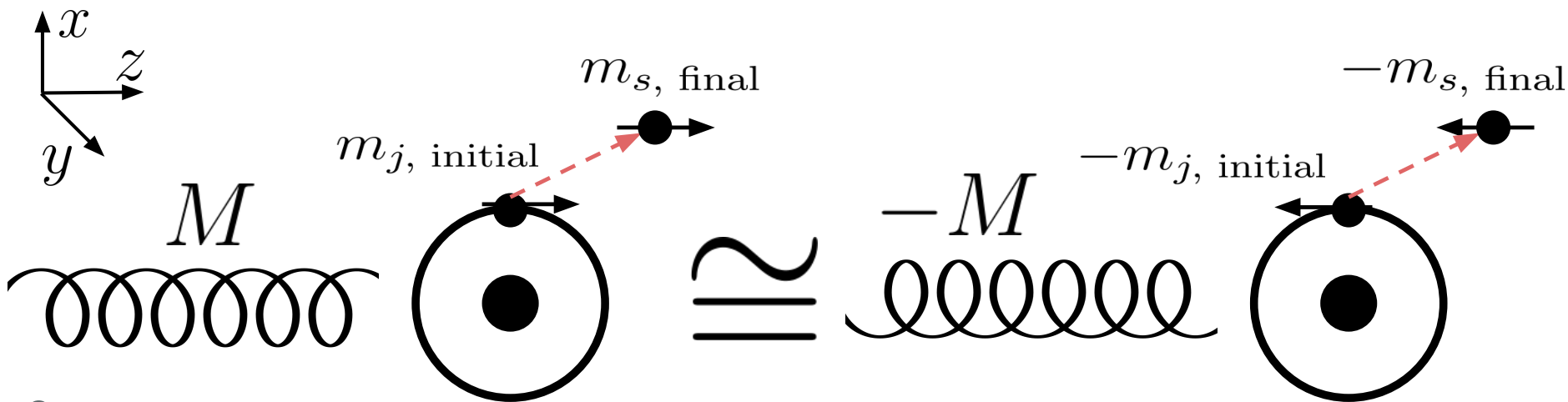
A. Kheifets et al. PRA 2016

- Kheifets discusses angle-resolved time delays for Argon 3p and Krypton 4p orbitals.
- Minimal difference of time delays between photoelectrons from 3p_{1/2} and 3p_{3/2}



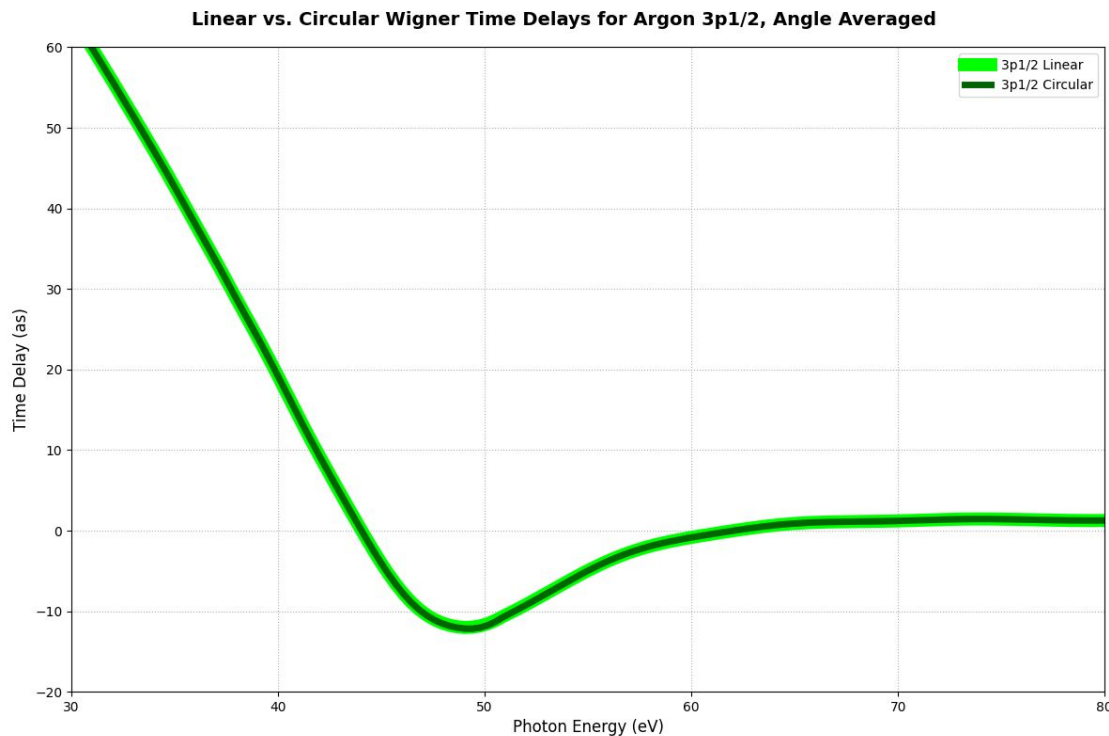
Circular Polarization

- System's axis of symmetry is the z-axis, preferred direction of spin basis
- Symmetry found in linear case is broken, allowing for spin-resolved differences
- A more general symmetry that allows for right-handed amplitudes to be turned into left-handed ones



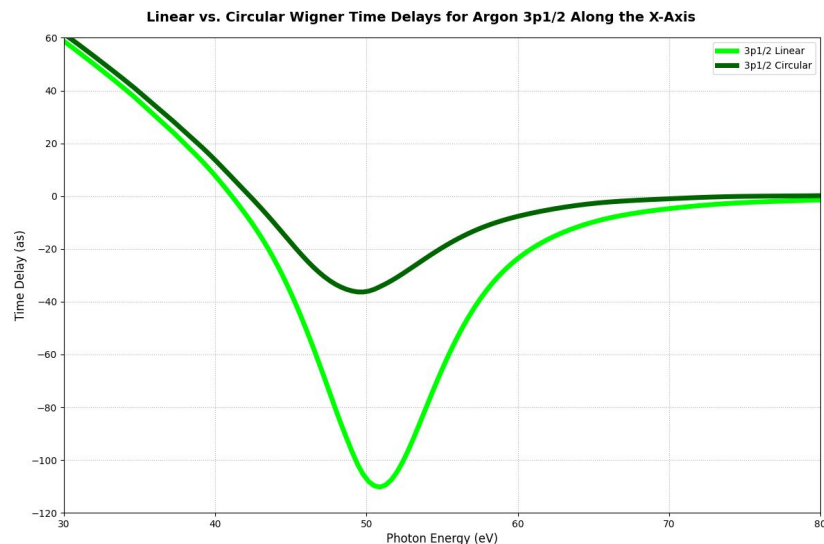
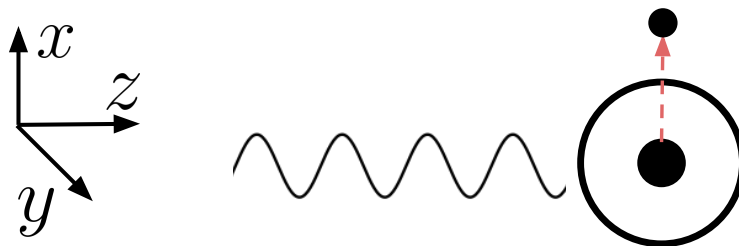
Differences Between Linear and Circular Delays

- Angular averaged time delays don't show a difference between linear and circular light.



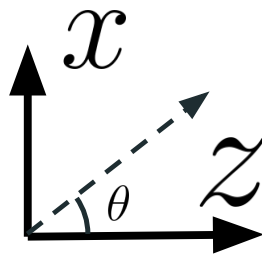
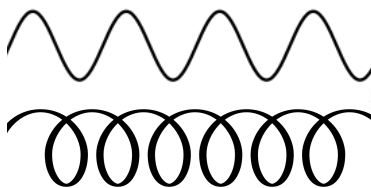
Differences Between Linear and Circular Delays

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- However, different geometries lead to different angular dependencies.

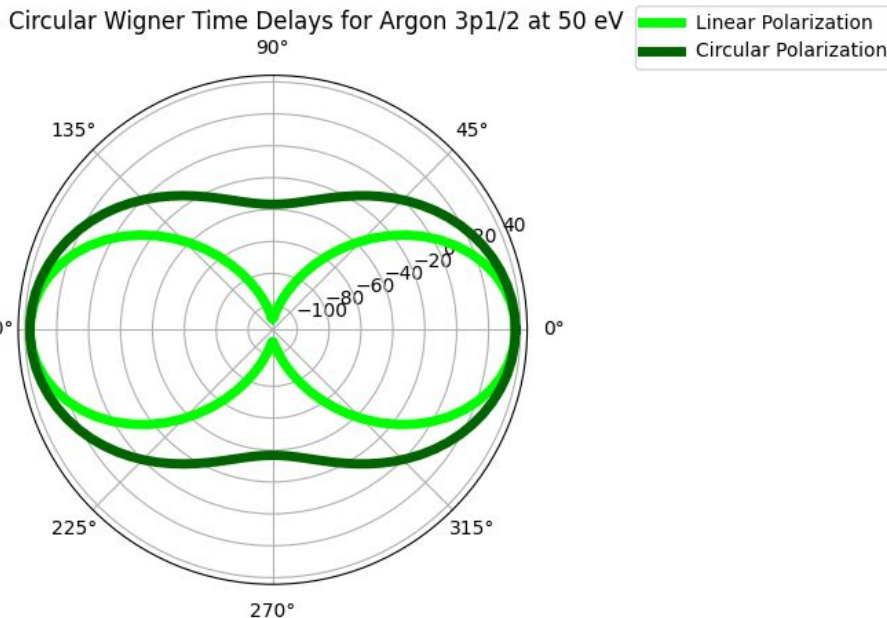


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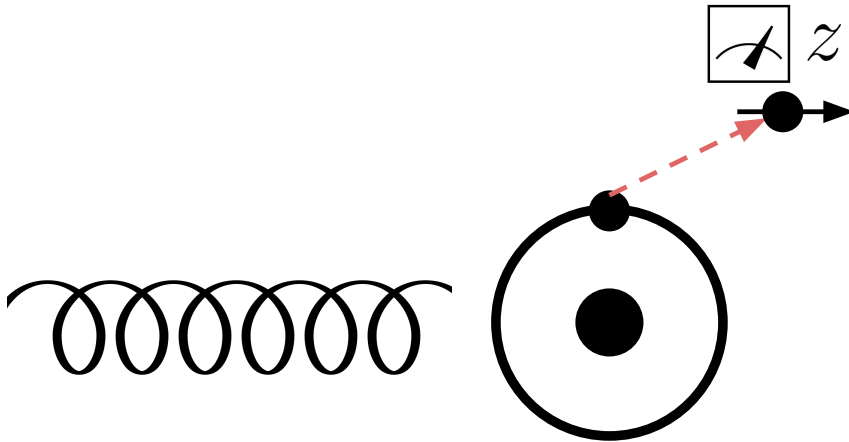


Linear vs. Circular Wigner Time Delays for Argon 3p_{1/2} at 50 eV

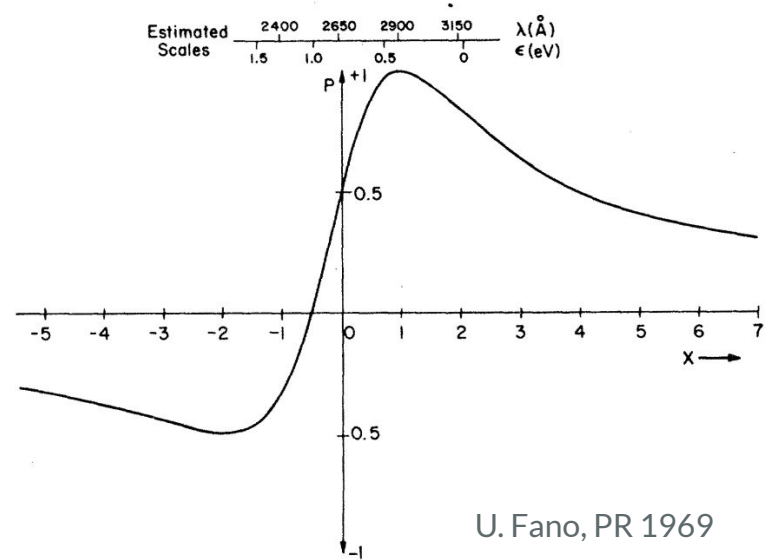


Spin Polarization

- Fano predicted that circularly polarized light would polarize photoelectrons from Cs 6s.
- Measuring spin along z-axis, direction of propagation



$$\mathcal{P} = \frac{\sigma_+ - \sigma_-}{\sigma_+ + \sigma_-}$$



U. Fano, PR 1969

Spin Polarization Results

- Overall polarization in s-orbital around the Cooper minima
- p_{1/2} and p_{3/2} suborbitals are polarized, but the overall p-orbital is unpolarized

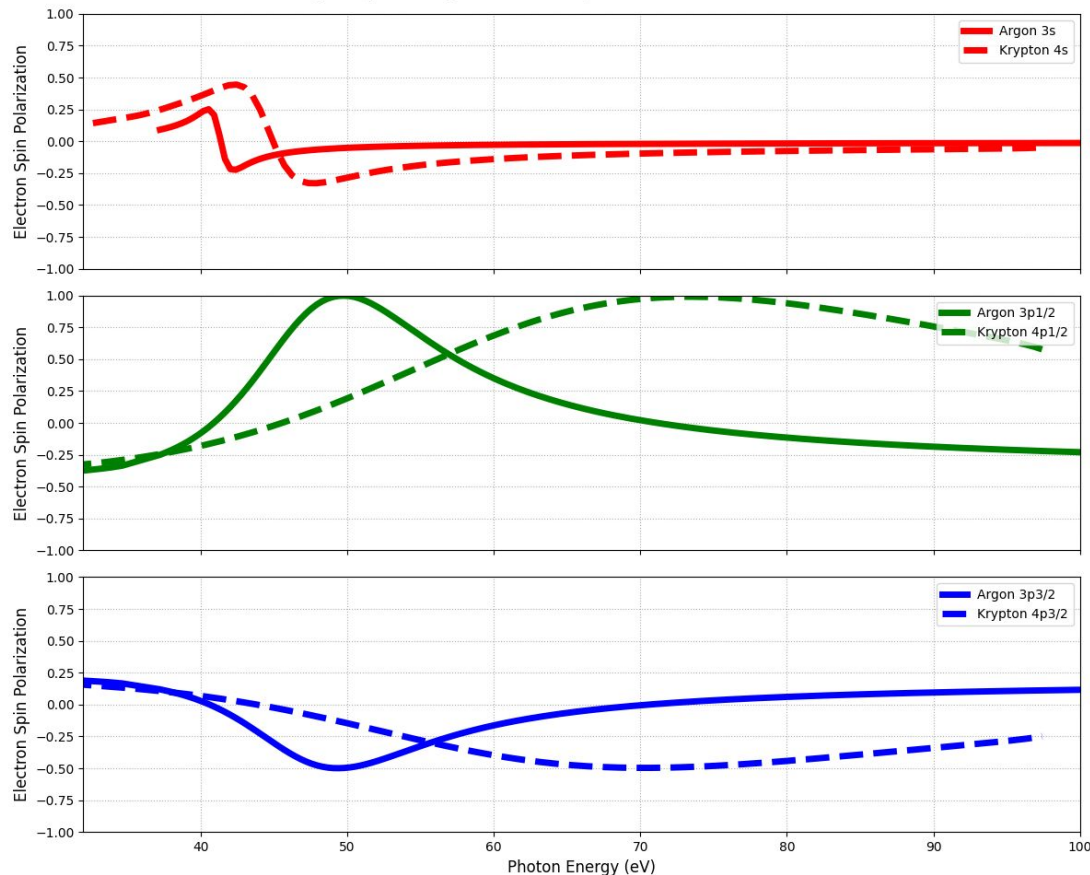
S

p_{1/2}

p_{3/2}

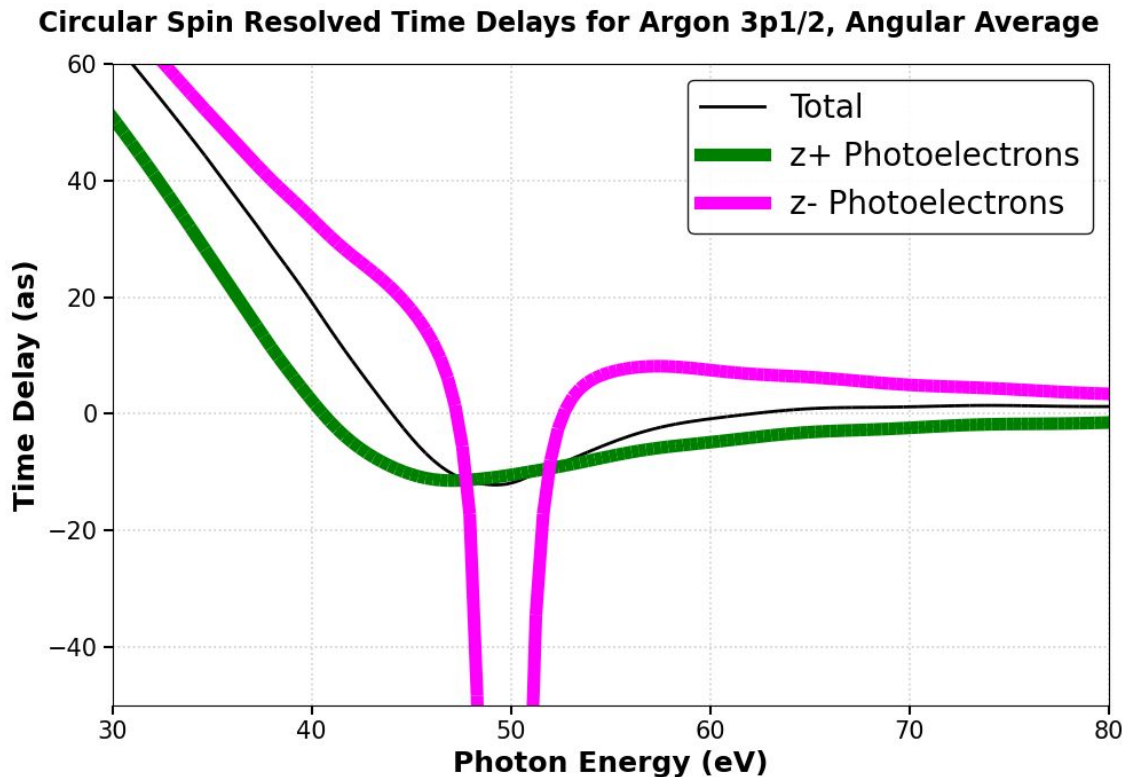
Argon — Krypton - - -

Angularly Averaged Circular Spin Polarization for Ar and Kr



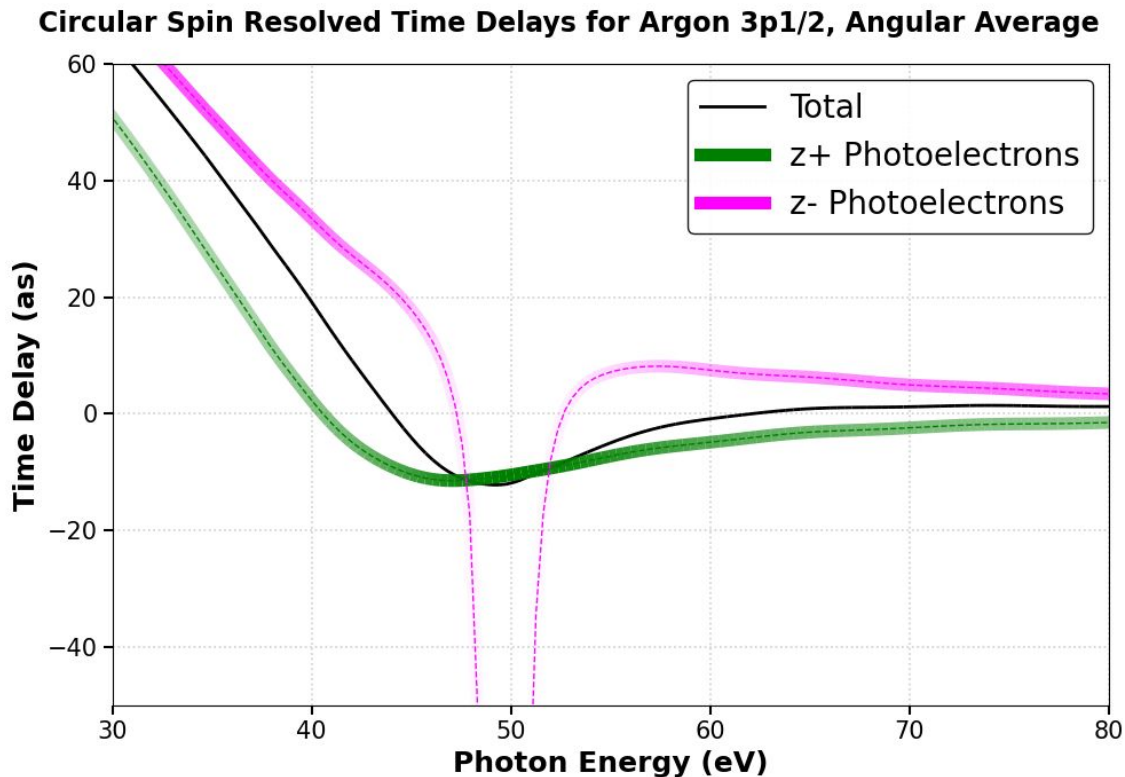
Spin-Resolved Time Delays

- Significant difference at Cooper minimum
- z- delay reaches -300 attoseconds.



Spin-Resolved Time Delays

- Significant difference at Cooper minimum
- z- delay reaches -300 attoseconds.
- However, the proportion of z- photoelectrons at this energy is zero.



Spin-Resolved Time Delays

- $p_{1/2}$ and $p_{3/2}$ separate in opposite directions, and the overall p-orbital shows no spin-resolved differences in time delay
- s-orbital shows noticeable differences at the Cooper minimum.

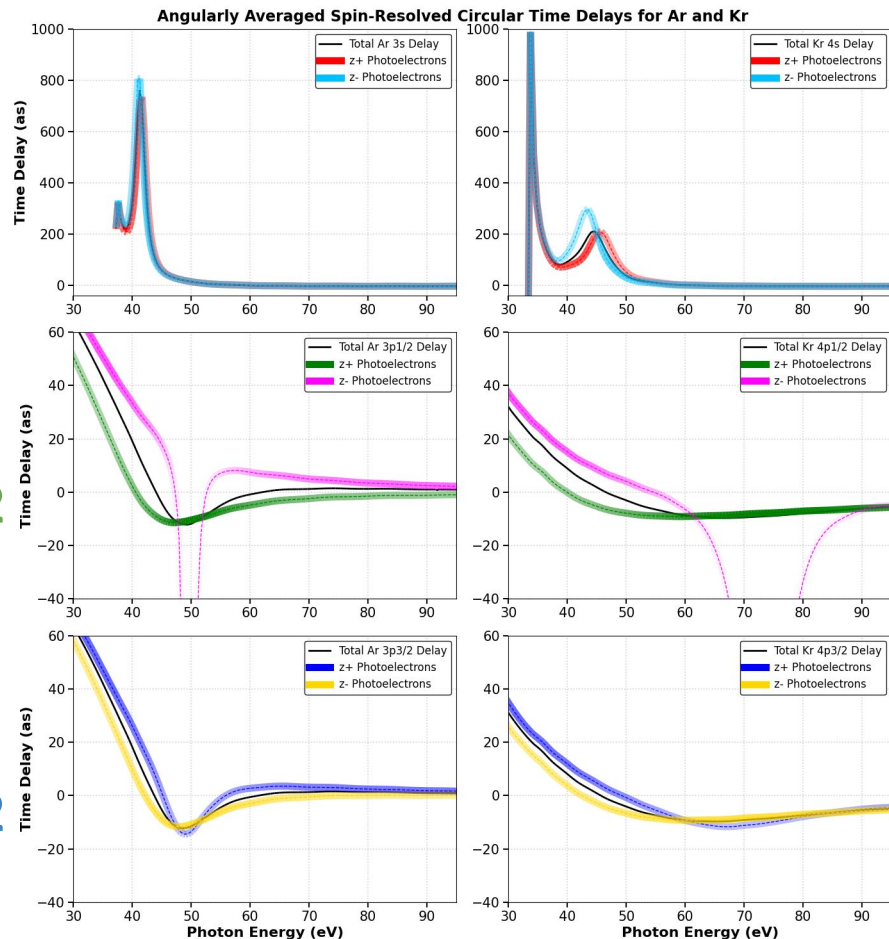
s

$p_{1/2}$

$p_{3/2}$

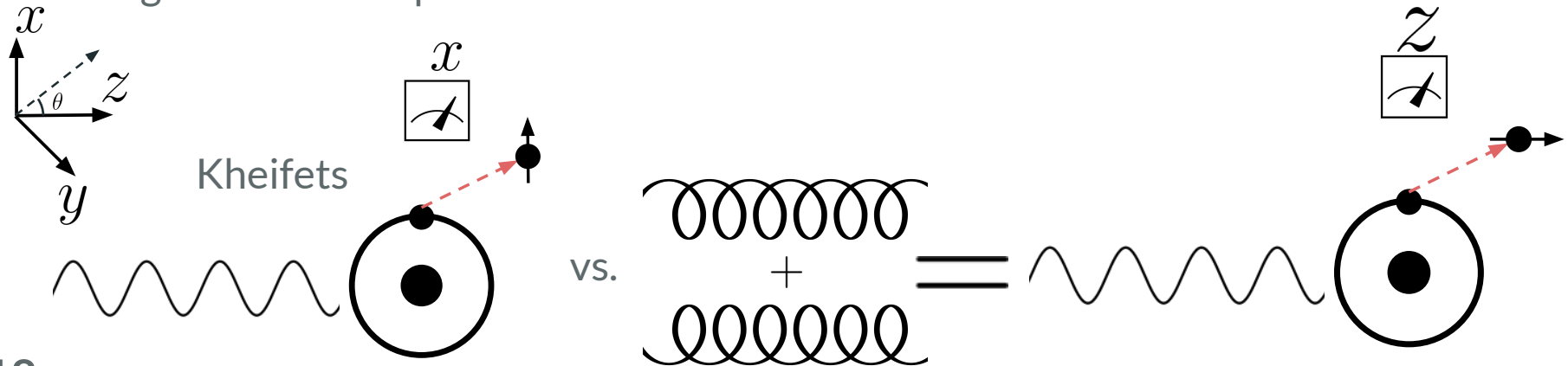
Argon

Krypton



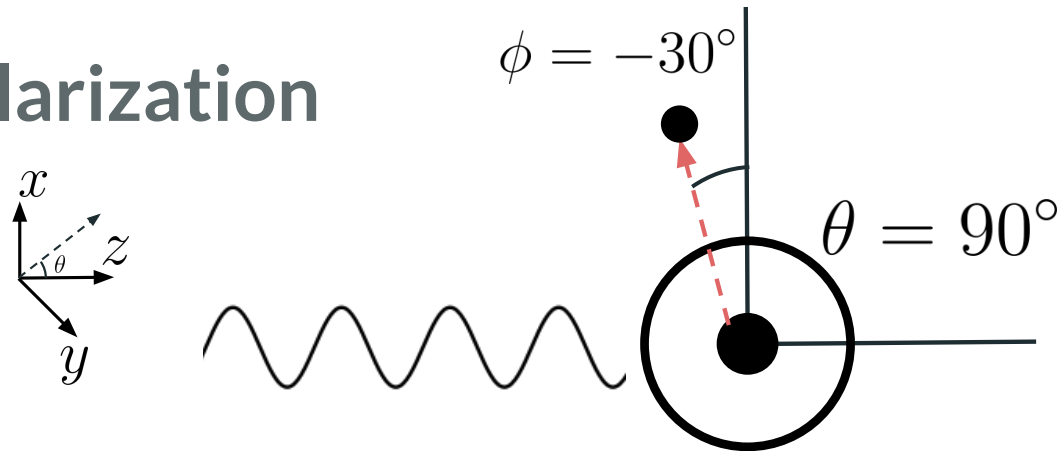
Revisiting Linear Polarization

- Kheifets paper looks at spin along the x-axis, and I have been looking at spin along the z-axis.
- For a more direct comparison using the z-axis basis, linear delays can be calculated using the sum of left and right-handed amplitudes

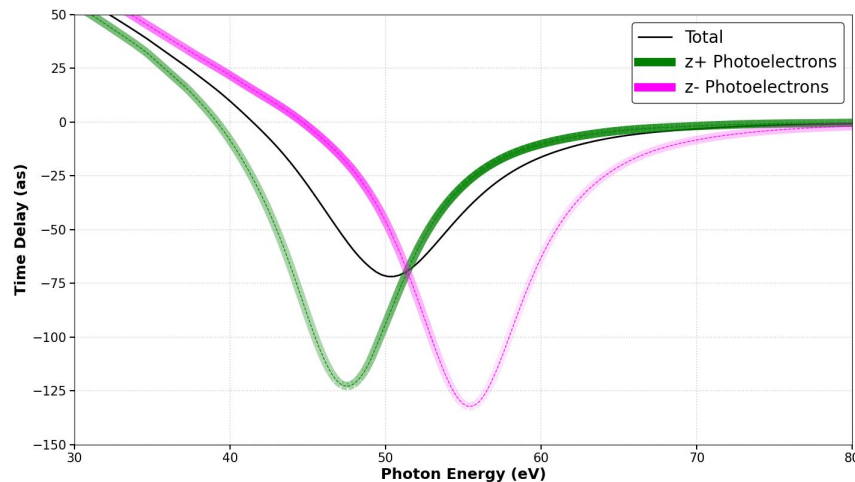


Revisiting Linear Polarization

- Angle-averaged spin-resolved delays are the same.
- However, at specific angles, z+ and z- delays have notable differences, with minima separating.

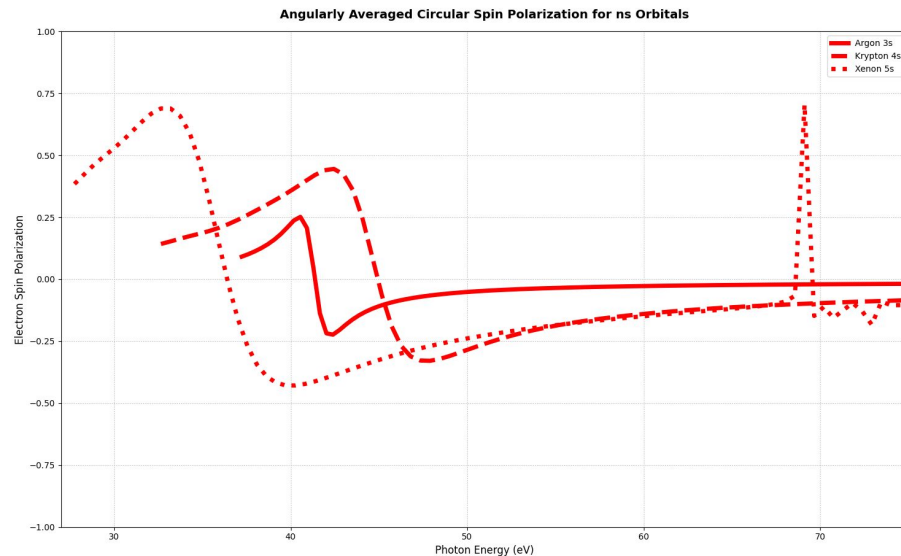
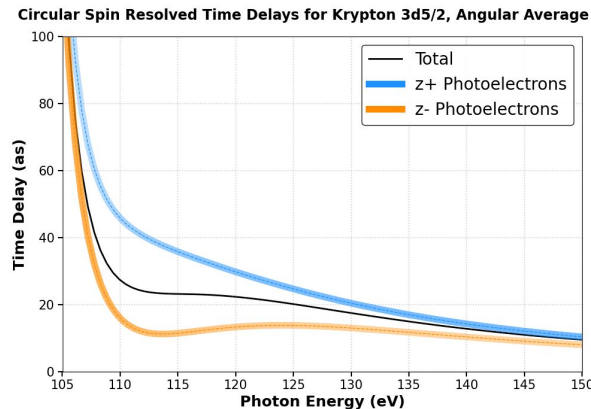
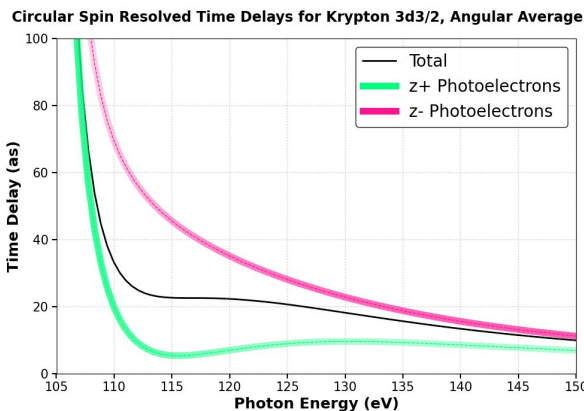


Linear Spin Resolved Linear Time Delays for Argon 3p1/2, at $\theta = 90^\circ$, $\phi = -30^\circ$



Future Work

- Look into higher orbitals, cancellation effect
- Look into Xenon, possibly Radon, periodic trends
- What do these results say about spin contributions to the potential?



Conclusions

- When angle-averaged, circular Wigner time delays are the same as their linear counterparts, but different geometries lead to different angular differences appear at specific angles.
- Spin polarization and spin-resolved circular delays reveal notable differences in the behavior of $p_{1/2}$ & $p_{3/2}$ suborbitals, which cancel when looking at the whole p-orbital.
- No such cancellation exists in the s-orbital.
- When measuring spin along the axis of propagation, angle-resolved linear time delays can be spin-dependent, with certain angles showing dramatic differences.